

Assignment Title:

Real-World Application of Stack Using Array-Based and Pointer-Based Implementation in C++

Assignment Instructions:

Group Assignment (4-5 Members) same group with Assignment 1

You are required to work in groups to explore the concept of **Stack** and its real-world applications. Follow the steps below to complete this assignment:

Task 1: Understanding the Concept of Stack

1. Explain the basic operations of a Stack (**Push, Pop, stackTop, IsEmpty, IsFull**) with examples (in C++ comments).
 2. Explain the **LIFO (Last-In-First-Out)** principle and its relevance in real-world scenarios (in reports).
-

Task 2: Application of Stack in a Real-World Scenario

1. Identify and choose **one real-world application** where the concept of Stack is used. Examples include:
 - Undo/Redo functionality in text editors.
 - Backtracking in algorithms (e.g., maze solving or navigating web browser history).
 - Evaluating mathematical expressions (Infix to Postfix conversion).
 - Postfix Calculator
 2. Provide a brief explanation of how the Stack data structure supports this application.
-

Task 3: Implementation in C++

1. Implement the chosen real-world application using the Stack data structure in **two ways**:
 - **Array-Based Implementation:** Use a fixed-size array to implement the stack.
 - **Pointer-Based Implementation (Linked List):** Use dynamic memory allocation and pointers to implement the stack.
2. Your implementation must include the following operations:
 - **Push:** Add an element to the stack.
 - **Pop:** Remove an element from the stack.
 - **stackTop:** Display the top element of the stack.
 - **Display:** Show all elements currently in the stack.

- **isFull and isEmpty** : when is a needed
3. Write a **menu-driven program** in C++ that allows users to perform the operations interactively.
-

Task 4: Comparative Analysis/Findings

1. Compare the **Array-Based** and **Pointer-Based** implementations in terms of:
 - Memory usage.
 - Ease of implementation.
 - Scalability.
 2. Write your findings in a summary report (200-300 words).
-

Deliverables:

1. **Program Code:** Submit the source code files (C++) for both implementations.
 2. **Report:** Include the following in your report or as a comment:
 - Explanation of the chosen real-world application.
 - Description of the stack operations.
 - Comparative analysis of the implementations.
 - Screenshots of program output for both implementations.
 3. Only the group leader should submit Assignment II. Please include the names of all group members on the front page.
-

Assessment Criteria:

- Understanding of Stack concepts (2%)
 - Relevance and explanation of the chosen real-world application (3%)
 - Correctness of the C++ implementations (5%)
 - Comparative analysis and report quality (5%)
-

Submission Deadline:

[1th January 2025] – Wednesday